

Road Network Annotation Description

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1. Course Contents

1. Use the route_tool tool to annotate `.geojson` format files describing road networks

2. Prerequisites

2.1 Start Agent

- Start the microROS agent and connect to the chassis. If already started, no need to restart

```
sh start_agent.sh
```

```
jetson@yahboom: ~  
jetson@yahboom: ~ 80x24  
subscriber created | client_key: 0x14F22590, subscriber_id: 0x000(4), participant_id: 0x000(1)  
[1763535321.445792] info | ProxyClient.cpp | create_datareader | d  
datareader created | client_key: 0x14F22590, datareader_id: 0x000(6), subscriber_id: 0x000(4)  
[1763535321.449556] info | ProxyClient.cpp | create_topic | t  
topic created | client_key: 0x14F22590, topic_id: 0x006(2), participant_id: 0x000(1)  
[1763535321.452876] info | ProxyClient.cpp | create_subscriber | s  
subscriber created | client_key: 0x14F22590, subscriber_id: 0x001(4), participant_id: 0x000(1)  
[1763535321.457712] info | ProxyClient.cpp | create_datareader | d  
datareader created | client_key: 0x14F22590, datareader_id: 0x001(6), subscriber_id: 0x001(4)  
[1763535321.461959] info | ProxyClient.cpp | create_topic | t  
topic created | client_key: 0x14F22590, topic_id: 0x007(2), participant_id: 0x000(1)  
[1763535321.466141] info | ProxyClient.cpp | create_subscriber | s  
subscriber created | client_key: 0x14F22590, subscriber_id: 0x002(4), participant_id: 0x000(1)  
[1763535321.473112] info | ProxyClient.cpp | create_datareader | d  
datareader created | client_key: 0x14F22590, datareader_id: 0x002(6), subscriber_id: 0x002(4)
```

2.2 Modify DDS

- Modify the ROS communication middleware DDS to use cyclonedds. Only modify when using road network features

roadnet DDS on

```
pi@raspberrypi:~ $ roadnet DDS on  
✓ Added to bashrc:  
export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp
```

- Check DDS status

roadnet DDS status

```
pi@raspberrypi:~ $ roadnet DDS status  
● cyclonedds is ON  
pi@raspberrypi:~ $
```

Refresh Environment Variables

- Orin board

```
source ~/.bashrc
```

- Raspberry Pi board, restart container

```
shut_m3pro  
bringup_m3pro
```

2.3 Start Road Network Container

```
roadnet start
```

```
jetson@yahboom:~$ roadnet start  
access control disabled, clients can connect from any host  
jetson@yahboom:~$
```

3. Road Network Annotation

- When annotating road networks, we need to know the exact position of the vehicle on the map. Therefore, we need to start the navigation node on the host machine first. The position can be known through the base_footprint coordinate system (PI5 users should start in the `m3pro container`)

```
ros2 launch M3Pro_navigation toolbox_location_nav.launch.py gui:=False
```

Note

Startup Parameter Description

- **gui** Whether to start rviz. Here we choose not to start; it will be started in the docker container later

3.1 Start Road Network Annotation Node

- Start in the roadnet container terminal

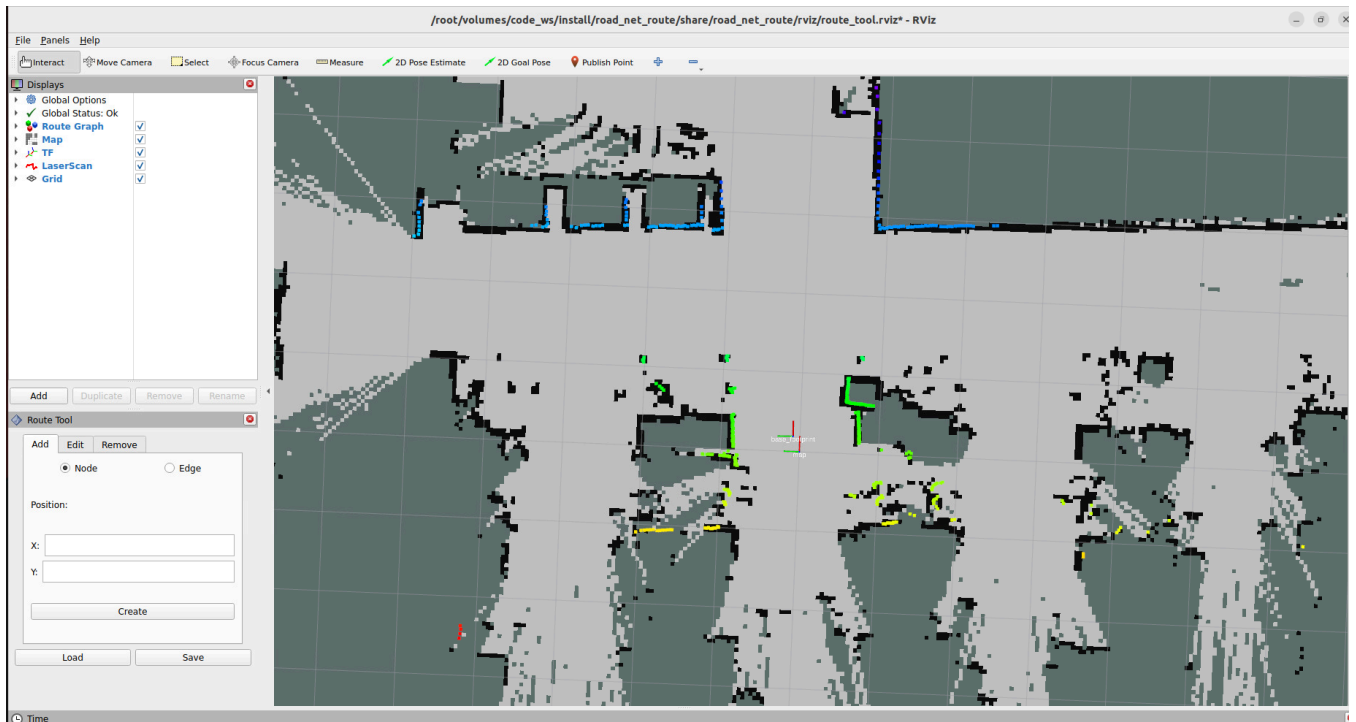
```
roadnet bash
```

```
ros2 launch road_net_route route_tool.launch.py yaml_filename:=map.yaml
```

Note

The map on the host machine is saved by default in the `~/map` path, which is automatically mounted to the `/root/map` directory inside the roadnet container. Therefore, map files saved on the host machine can be used directly in the roadnet container

- In rviz, the map and two coordinate frames will be displayed. The map is the map coordinate frame, and base_footprint is the coordinate frame of the projection point of the vehicle chassis center on the ground. Through base_footprint, we can know the exact position of the vehicle on the map in real-time

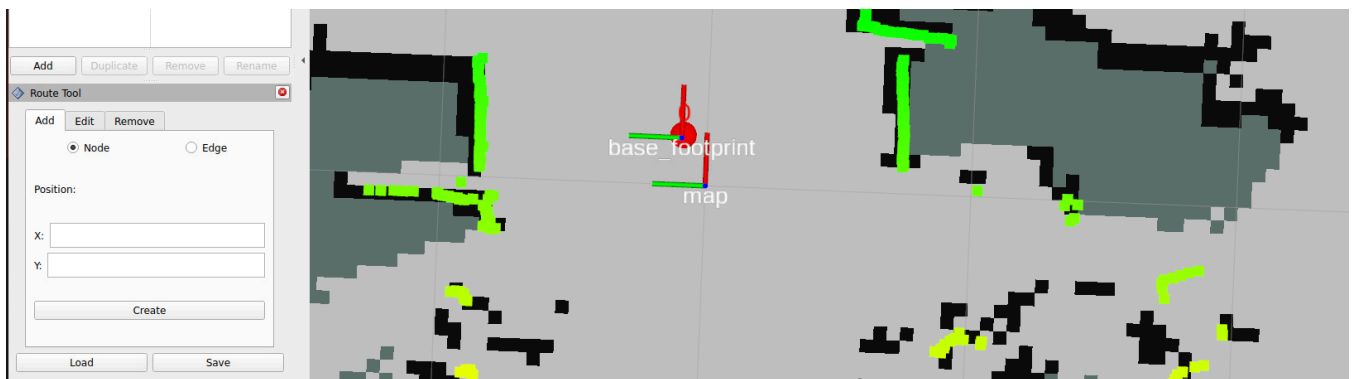
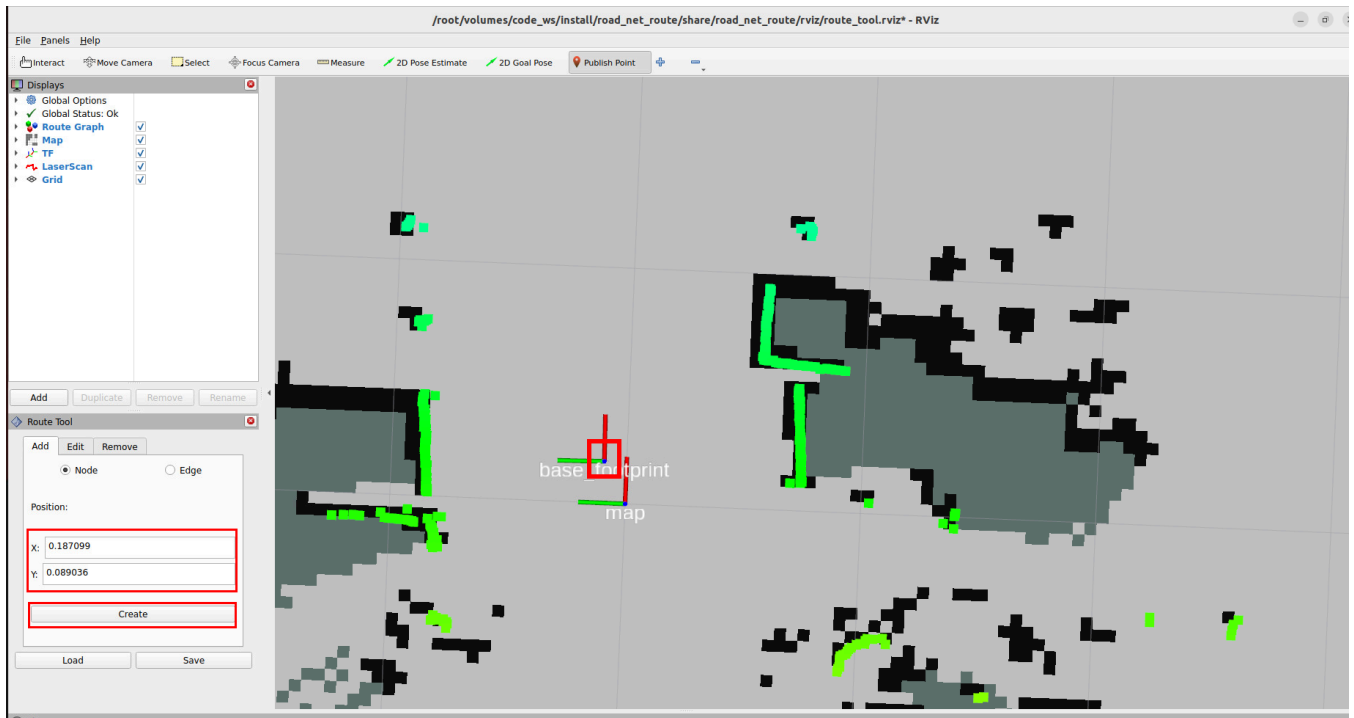


3.2 Add Waypoints

- Click the `Publish Point` tool in the rviz toolbar to enter waypoint selection mode



- Then click the left mouse button on the blue dot at the center of the `base_footprint` coordinate frame. The `position` in the `Route Tool` tab at the bottom left will automatically fill in the current point's coordinates. Then click `Create` to create a waypoint

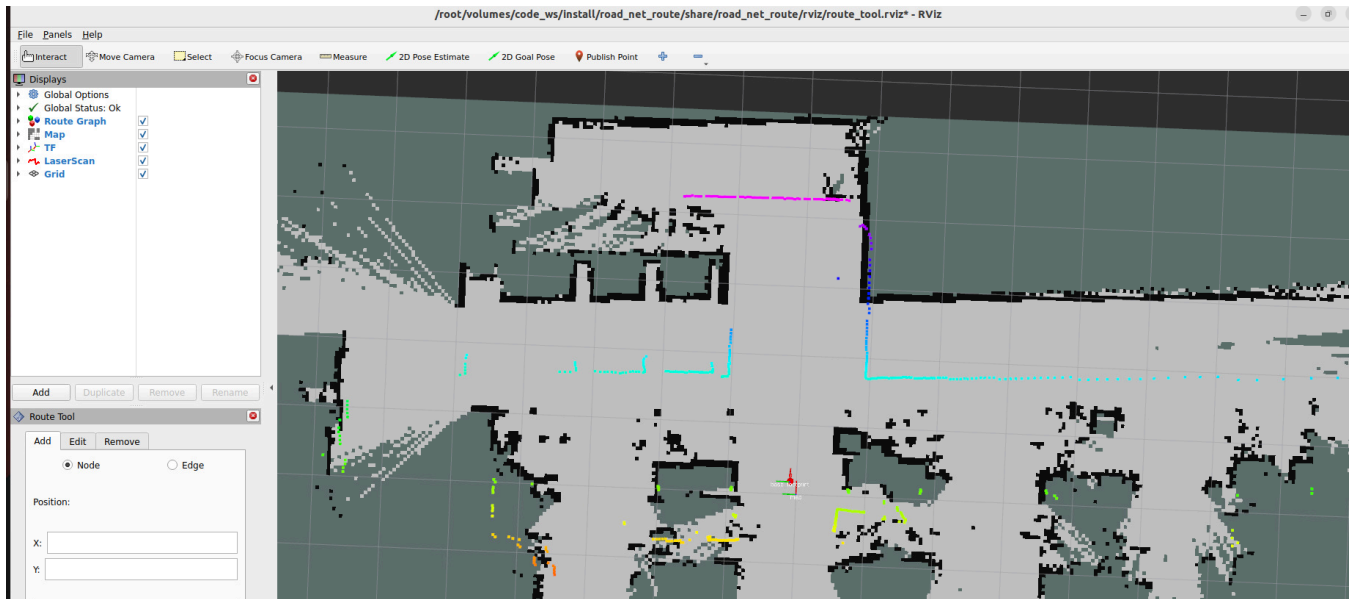


- Next, move the vehicle to the next position where a waypoint needs to be annotated. In this example, the vehicle is directly moved to the next position where a waypoint needs to be marked

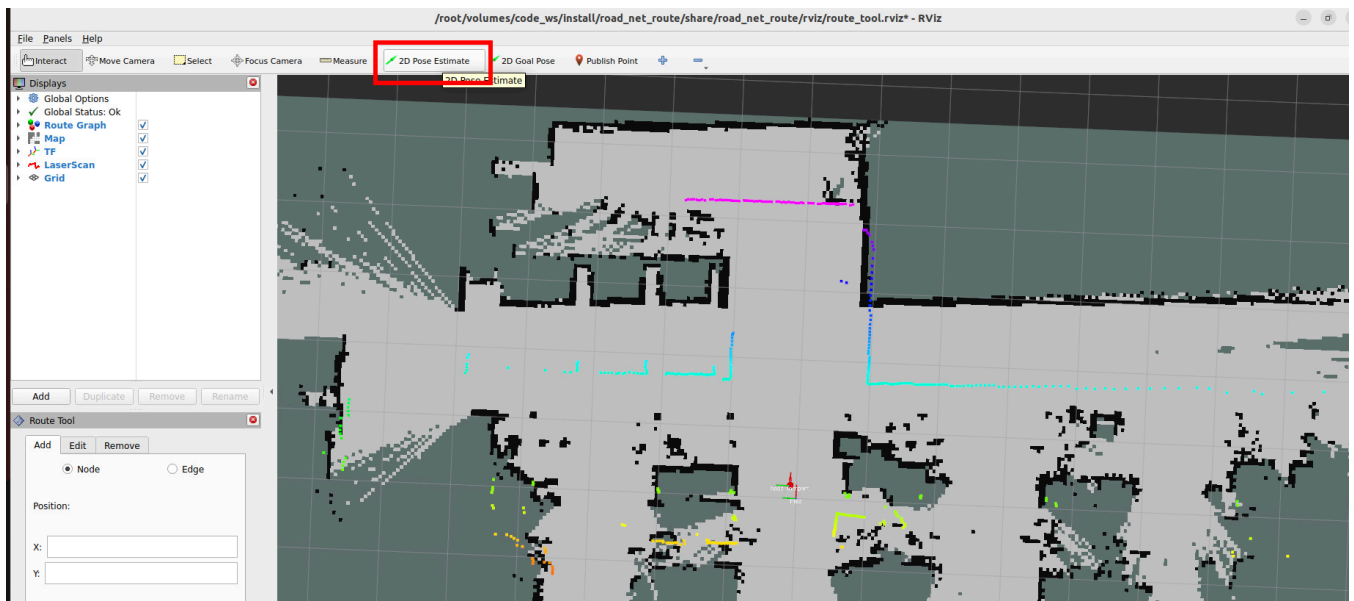
💡 Tip

The vehicle can be moved through keyboard control, gamepad control, `2D Goal Pose` navigation in rviz, or even by directly physically moving the vehicle to the next position where a waypoint needs to be marked. Since there is a pose map, the vehicle's global position can be quickly and accurately located

- As you can see, directly moving the vehicle causes the global localization to be lost, and the laser outline no longer matches the map environmental features. Here we can use the pose map for rapid relocalization



- Click the **2D Pose Estimate** tool in the rviz toolbar



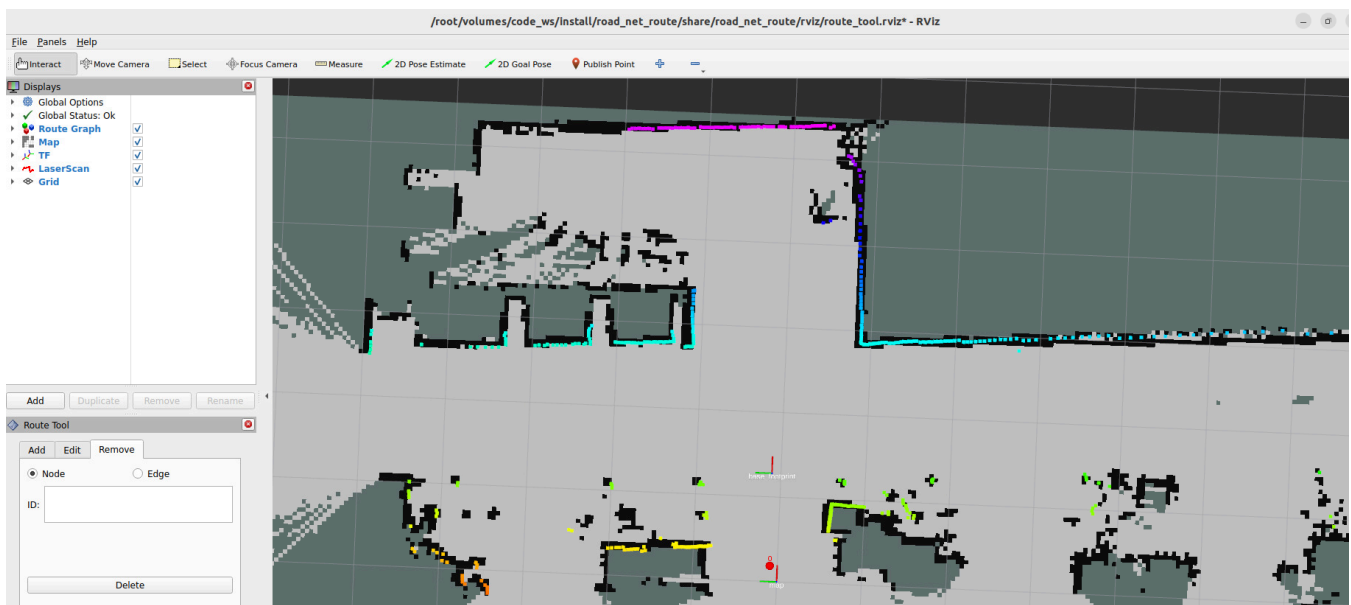
- Then provide an estimated position near the approximate location of the vehicle



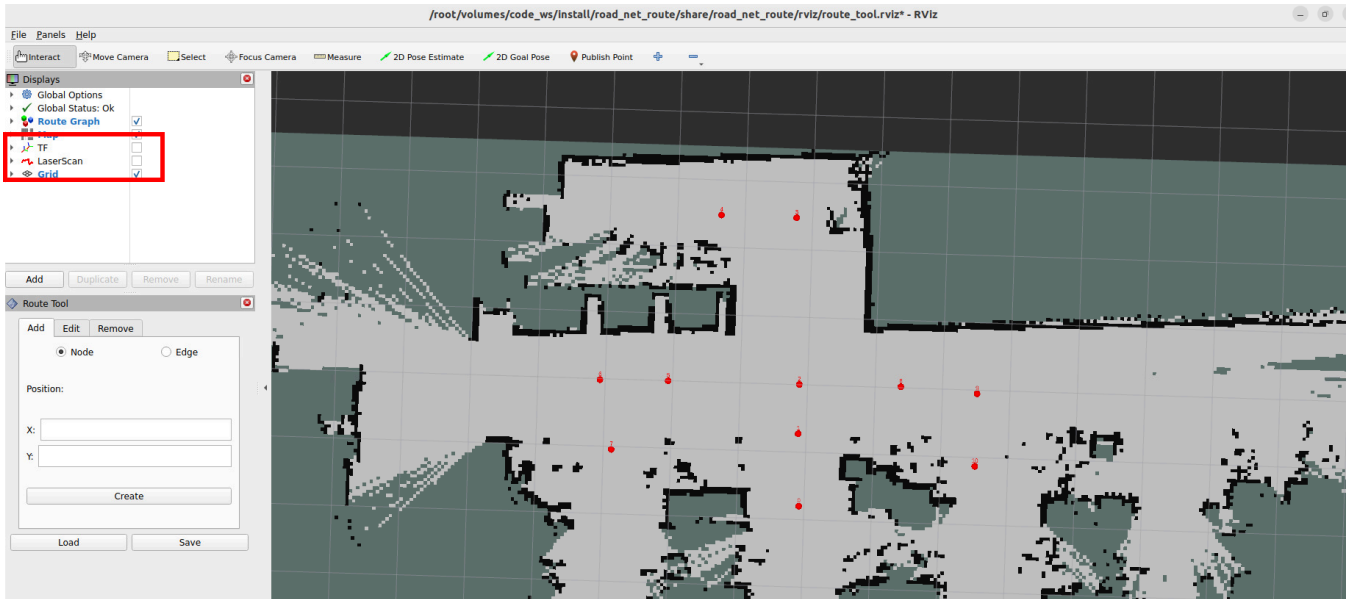
- After this, the vehicle's accurate position in the global map will be immediately restored

💡 Tip

- The rapid relocalization principle here is to use slam_toolbox pure localization and pose map for rapid scan-map matching
- **Note:** When observing **deviation between the laser outline and map environmental features** during subsequent waypoint annotation (due to cumulative odometry errors, wheel slippage, manual movement, or other factors causing local localization errors), this method can be used to quickly restore the vehicle's accurate position.



- Repeat the above steps until all waypoints that need to be annotated are completed (annotate waypoints according to your actual needs)
- After annotation is complete, you can first close the left TF and LaserScan to inspect the waypoints



3.3 Add Edges

- Select Edge in the `Route Tool` tab



- Fill in the starting waypoint in `Start Node ID`, fill in the ending waypoint in `End Node ID`, and then click Create to create an edge between waypoints

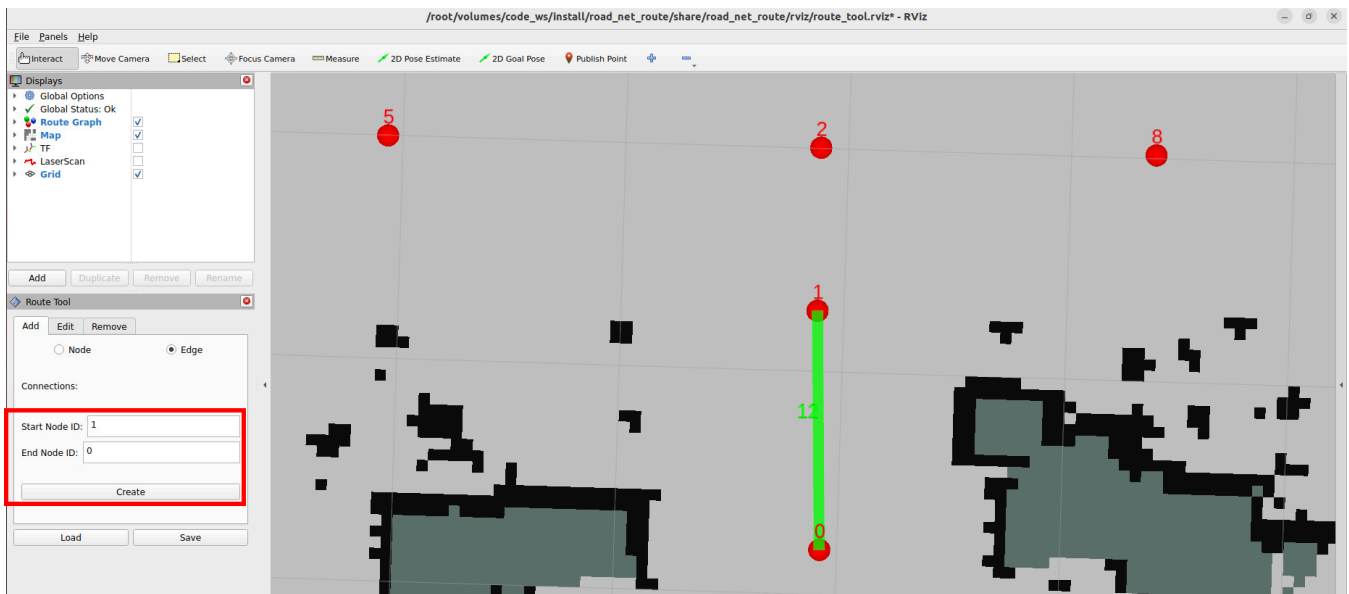
Important

Attention!!!

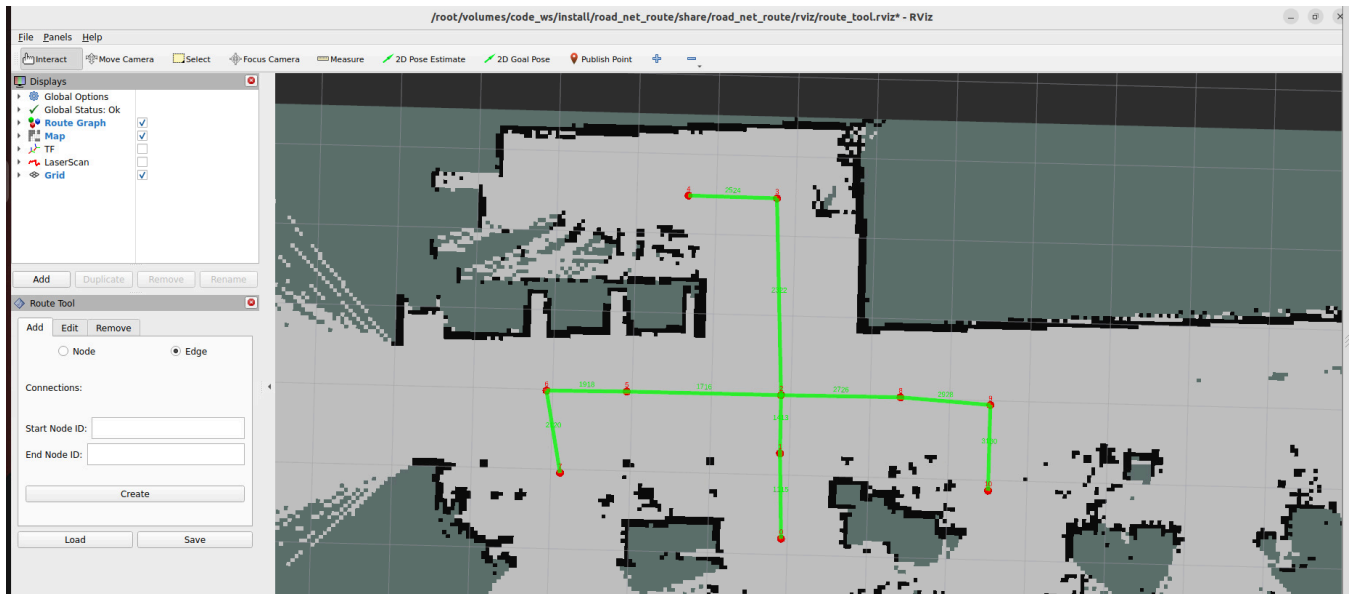
The road network is a topological structure composed of points and edges. Edges have direction. For example, creating an edge 0---->1 here is unidirectional. If bidirectional edges are needed, you also need to create edge 1---->0.



- Next, we create the 1-0 edge so that waypoints 0 and 1 can travel bidirectionally

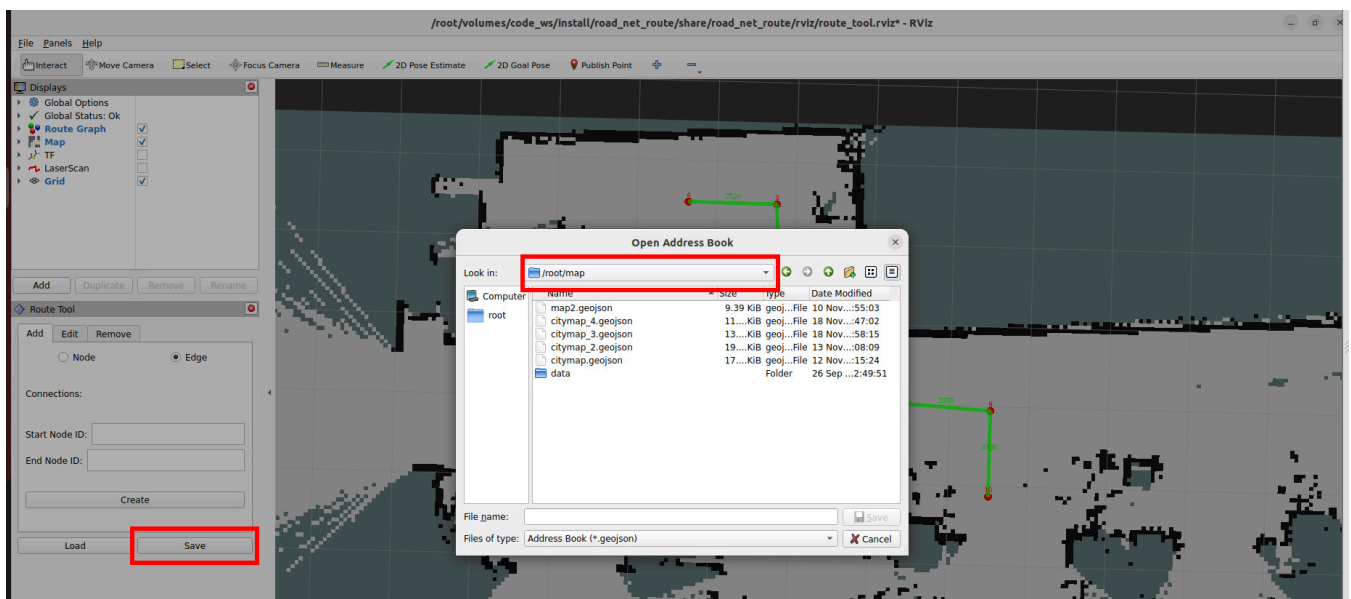


- Repeat the above steps to complete all edge creation (whether to create unidirectional or bidirectional edges depends on the actual scenario requirements)

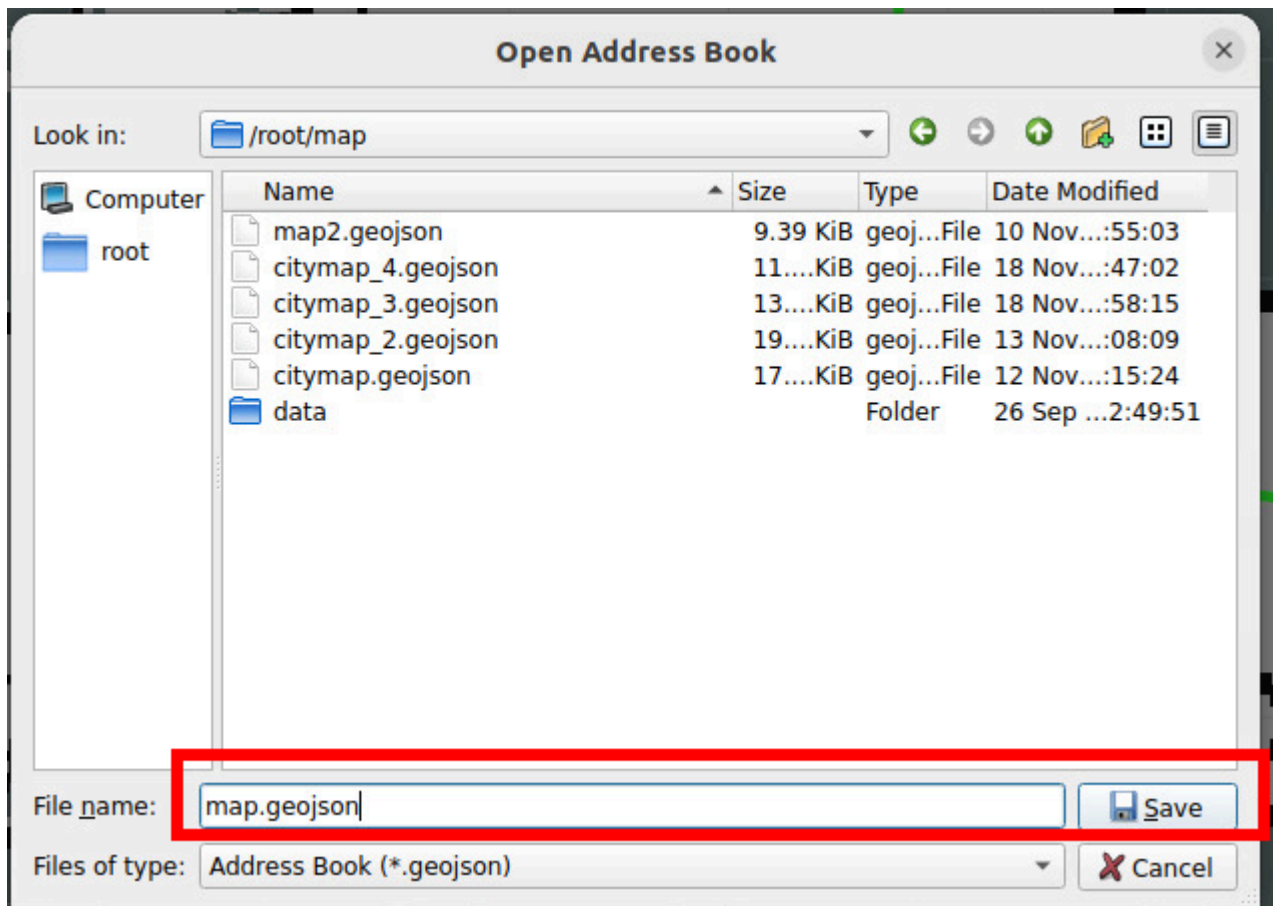


4. Save Road Network Description File

- Click save in the `Route Tool` tab, then select the file path in the pop-up window. Here we select the `/root/map` path because this path is bound to the host machine's `~/map` path. Later, you can directly view the saved file on the host machine



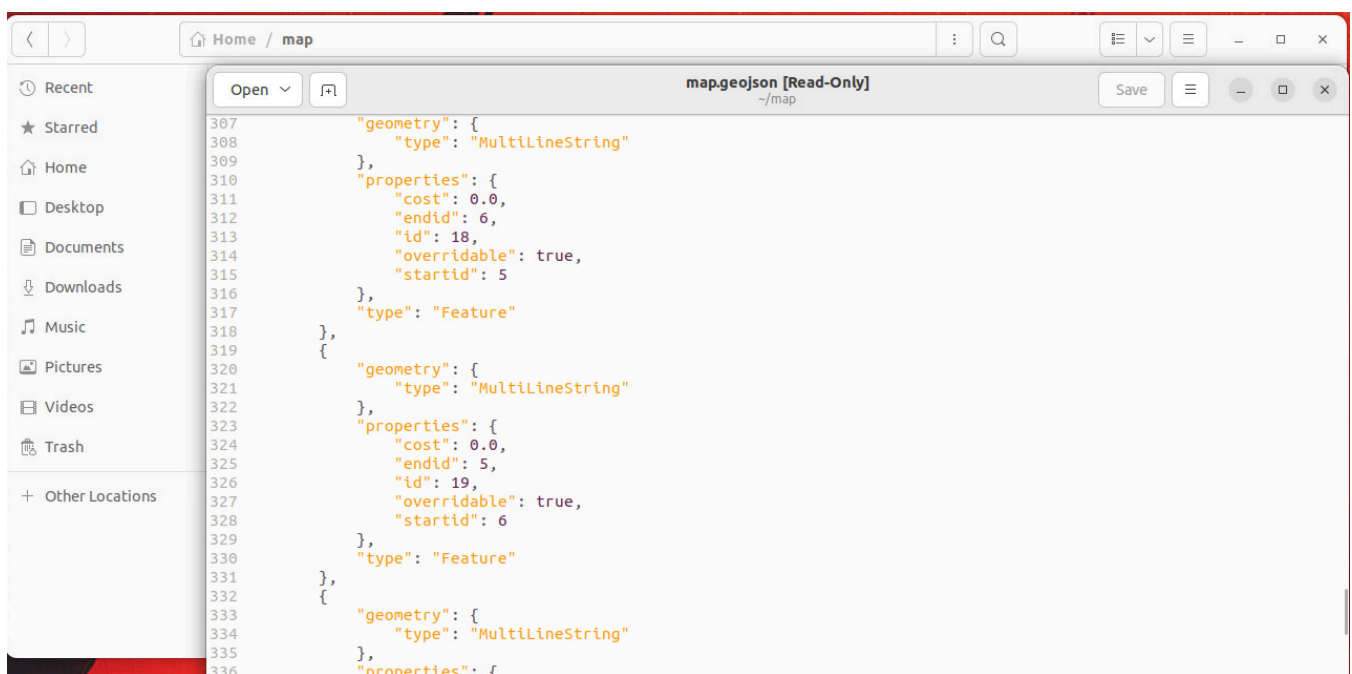
- Then fill in the file name in File name. It is recommended to keep the same name as the raster map for easy identification. The file suffix must be **.geojson**. Finally, click save to save the road network description file



- After successful save, the terminal will print a prompt

```
[rvt2-1] [INFO] [1763543213.88141103] [route_tool_node]: Save graph to:
[rvt2-1] [WARN] [1763543213.881559940] [route_tool_node]: The graph filepath was not provided and no default was specified. Failed to save the route graph.
[rvt2-1] [INFO] [1763543726.803456606] [route_tool_node]: Save graph to: /root/map/map.geojson
[rvt2-1] [INFO] [1763543726.803604931] [route_tool_node]: Saving graph file from /root/map/map.geojson, by parser nav2_route::GeoJsonGraphFileSaver
```

- Afterwards, we can directly view the map.geojson road network description file from the host machine's `~/map` path. The file contains a json format description of waypoint and edge information



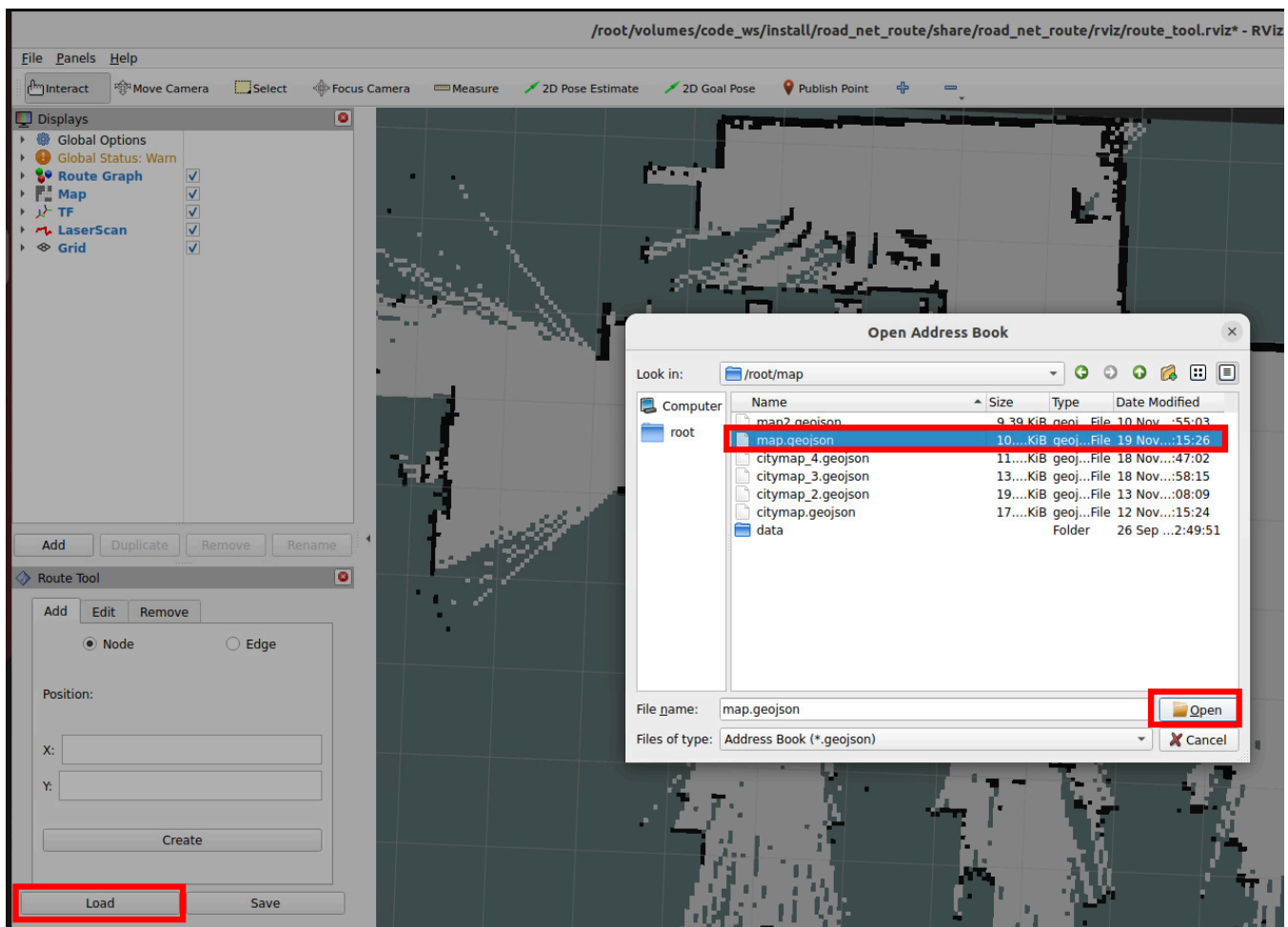
5. Load and Modify Road Network Description

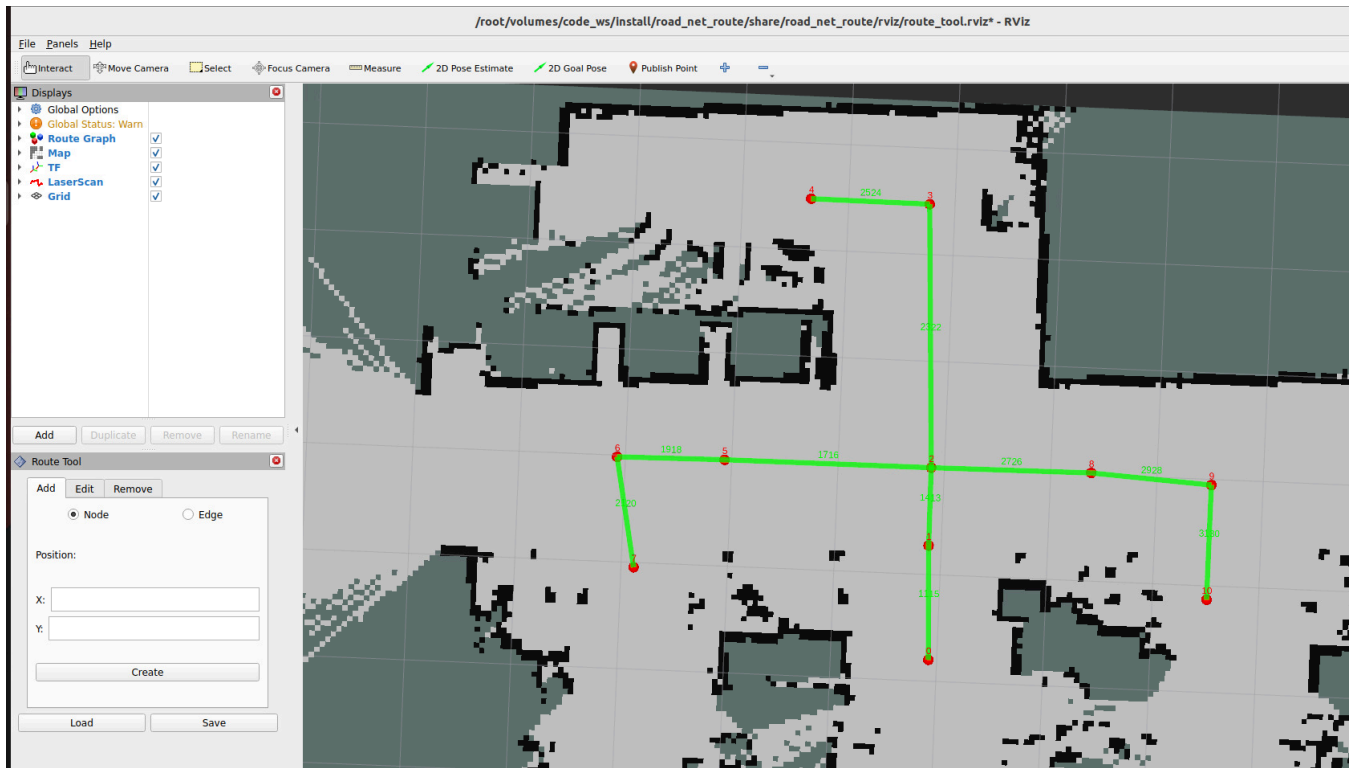
5.1 Load Road Network Description File

- If you need to modify a previous road network description file later, such as adding, removing, editing points or edges, you can load the previous road network description file for editing
- Start in the roadnet container terminal

```
ros2 launch road_net_route route_tool.launch.py yaml_filename:=map.yaml
```

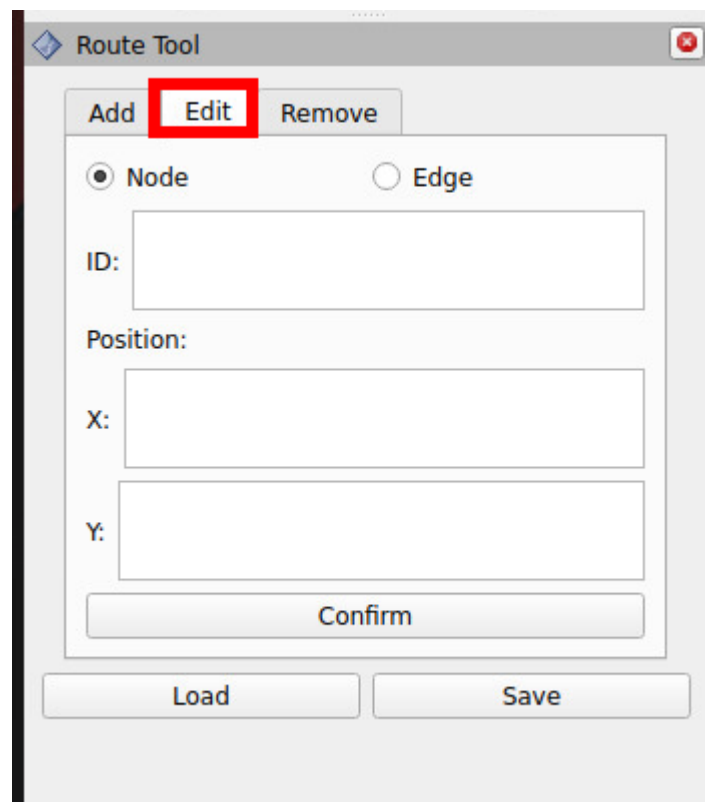
- **yaml_filename** Parameter is the raster map file path of the road network to be annotated





5.2 Edit Road Network

- If you need to add new waypoints and edges, the operation is the same as the previous steps
- Click **Edit** to re-edit information for existing waypoints or edges



- Click **Remove** to remove existing waypoints or edges

Route Tool

Add Edit **Remove**

☒ Node ☐ Edge

ID:

Delete

Load Save